Plane

Formulae used in Planes:

(1) The general equation of first degree in x, y, z represent a plane and its equation is given by

$$ax+by+cz+d=0$$

(2) The general equation of a plane that passes through the point (x_1, y_1, z_1) is given by

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$$

P.T.O

(4) The equation of a plane that passes through the points (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) is given by

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

(5) The equation of a plane which cuts intercepts a, b, c from the axes of reference is given by

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

← Equ^n of Plane: Intercept form

(5) The equation of a plane that passes through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by

$$(x-x_1)(y-y_2)-(x-x_2)(y-y_1)=k\{(x-x_1)(z-z_2)-(x-x_2)(z-z$$

(6) Equation of yz-plane: $x=0$
Equation of zx-plane: $y=0$
Equation of xy-plane: $z=0$

(7) Condition of the four points being coplanar is (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) & (x_4, y_4, z_4)

$$\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0$$

(8) If (l, m, n) be the dc^s of the normal of the plane and p be the length of that normal from the origin, then the equation of that plane is given by

$$lx+my+nz=p$$ ← Equ^n of Plane: (Normal form)

(9) Equation of the plane that intersects the two given planes $a_1x+b_1y+c_1z+d_1=0$ & $a_2x+b_2y+c_2z+d_2=0$ is given by

$$a_1x+b_1y+c_1z+d_1+k(a_2x+b_2y+c_2z+d_2)=0$$

(10) If θ be the angle between two planes $a_1x+b_1y+c_1z+d_1=0$ & $a_2x+b_2y+c_2z+d_2=0$, then

$$\boxed{\cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\cdot\sqrt{a_2^2+b_2^2+c_2^2}}}$$

Remark :-

(i) If the two planes represented by $a_1x+b_1y+c_1z+d_1=0$ & $a_2x+b_2y+c_2z+d_2=0$ are perpendicular to each other, then

$$\boxed{a_1a_2+b_1b_2+c_1c_2=0}$$

(ii) If the two given planes are parallel to each other, then

$$\boxed{\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}}$$

(11) The equation of the plane which is parallel to the given plane $ax+by+cz+d=0$, is given by

$$\boxed{ax+by+cz+K=0}$$

(12) Equation of the plane which passes through the point (x_1, y_1, z_1) and perpendicular to the planes $a_1x+b_1y+c_1z+d_1=0$ & $a_2x+b_2y+c_2z+d_2=0$ is given by

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

(13) The perpendicular distance from the point (x_1, y_1, z_1) to the plane $ax+by+cz+d=0$ is given by

$$\boxed{d = \frac{|a_1x+b_1y+c_1z+d|}{\sqrt{a^2+b^2+c^2}}}$$

(14) The equation of the two bisector planes of the angles between two planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ is given by

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \pm \frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$$

(15) The ~~general~~ equation of second degree - $ax^2+by^2+cz^2+2fyz+2gzx+2hxy=0$ represents a pair of planes if $\Delta=0$

i.e $\boxed{abc+2fgh-af^2-bg^2-ch^2=0}$

or, $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

(16) If the ~~general~~ equation of second degree $ax^2+by^2+cz^2+2fyz+2gzx+2hxy=0$ represents a pair of planes, then the angle between the planes is given by

$$\tan\theta = \frac{2\sqrt{f^2+g^2+h^2-bc-ca-ab}}{a+b+c}$$

Remark:

1) If the two planes are parallel to each other then the angle between them is $\theta = 0^\circ$, therefore

$$f^2 + g^2 + h^2 = bc + ca + ab$$

Plane Related Problems

Problem:

Reduce the equation of the plane $2x+3y-6z+9=0$ to the normal form and hence find the direction cosines of the normal to the plane.

Sol^n:

Given, $2x+3y-6z+9=0$ ——— ①

Now, $\sqrt{2^2+3^2+(-6)^2} = \sqrt{49} = 7$.

Diving ① by 7 throughout we get

$$\frac{2}{7}x+\frac{3}{7}y-\frac{6}{7}z+\frac{9}{7}=0$$

$$\Rightarrow -\frac{2}{7}x-\frac{3}{7}y+\frac{6}{7}z=\frac{9}{7} \quad \text{——— ②}$$

which is the required normal form of the eqn of the plane ①.

Comparing ② with $lx+my+nz=p$ we obtain

$$l=-\frac{2}{7},\ m=-\frac{3}{7},\ n=\frac{6}{7}.$$

(Ans)

Problem:

Find the equation of the plane that passes through the points $(2,1,-3)$, $(3,-1,4)$ & $(7,5,6)$.

Soln:

We know that the equation of the plane that passes through three points (x_1, y_1, z_1), (x_2, y_2, z_2) & (x_3, y_3, z_3) is given by

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

Thus,

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 1 & -3 & 1 \\ 3 & -1 & 4 & 1 \\ 7 & 5 & 6 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-2 & y-1 & z+3 & 0 \\ -1 & 2 & -7 & 0 \\ -4 & -6 & -2 & 0 \\ 7 & 5 & 6 & 1 \end{vmatrix} = 0$$

$r_1' = r_1 - r_2$

$r_2' = r_2 - r_3$

$r_3' = r_3 - r_4$

$$\Rightarrow \begin{vmatrix} x-2 & y-1 & z+3 \\ -1 & 2 & -7 \\ -4 & -6 & -2 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(-4-42) - (y-1)(2-28) + (z+3)(6+8) = 0$$

$$\Rightarrow -46(x-2) + 26(y-1) + 14(z+3) = 0$$

$$\Rightarrow 23(x-2) - 13(y-1) - 7(z+3) = 0$$

$$\Rightarrow 23x - 13y - 7z - 46 + 13 - 21 = 0$$

$$\Rightarrow 23x - 13y - 7z - 54 = 0$$

Ans

Problem:

Show that the four points $(3,2,5)$, $(0,1,3)$, $(5,4,2)$ and $(7,0,-1)$ are coplanar.

Sol^n:

We know that the four given points will be coplanar if

$$\begin{vmatrix} -3 & 2 & 5 & 1 \\ 0 & 1 & 3 & 1 \\ 5 & 4 & 2 & 1 \\ 7 & 0 & -1 & 1 \end{vmatrix} = 0$$

Now,

$$\begin{vmatrix} -3 & 2 & 5 & 1 \\ 0 & 1 & 3 & 1 \\ 5 & 4 & 2 & 1 \\ 7 & 0 & -1 & 1 \end{vmatrix} = -3 \begin{vmatrix} 1 & 3 & 1 \\ 4 & 2 & 1 \\ 0 & -1 & 1 \end{vmatrix}$$

$$-2 \begin{vmatrix} 0 & 3 & 1 \\ 5 & 2 & 1 \\ 7 & -1 & 1 \end{vmatrix} + 5 \begin{vmatrix} 0 & 1 & 1 \\ 5 & 4 & 1 \\ 7 & 0 & 1 \end{vmatrix}$$

$$-1 \begin{vmatrix} 0 & 1 & 3 \\ 5 & 4 & 2 \\ 7 & 0 & -1 \end{vmatrix}$$

$$= -3[3 - 3(4-0) + 1(-4-0)]$$

$$-2[0(2+1) - 3(5-7) + 1(-5-14)]$$

$$+5[0(4-0) - 1(5-7) + 1(0-28)]$$

$$-1[0(-4-0) - 1(-5-14) + 3(0-28)]$$

$$= -3(-13) - 2(-13) + 5(-26) - 1(-65)$$

$$= 0$$

Hence, the points are coplanar.

(Showed)

Problem:

Find the equation of the plane which passes through the point $(1,-5,-2)$ and whose normal has dr's $2,3,-7$.

Soln:

We know that the equation of the plane that passes through the point (x_1, y_1, z_1) is given by

$$a(x-x_1) + b(y-y_1) + c(z-z_1) \quad \text{——— (1)}$$

where a, b, c are the direction ratios of the normal to the plane.

Thus, we have from (1)

$$2(x-1) + 3(y+5) - 7(z+2) = 0$$

$$\Rightarrow 2x + 3y - 7z - 2 + 15 - 14 = 0$$

$$\Rightarrow 2x + 3y - 7z - 1 = 0$$

02.02.2023

Ans

Problem:

Find the equation of the plane that passes through the origin and is normal to the line joining the points $(5,3,-1)$ and $(-2,-6,4)$.

Soln:

We know that the eqn of the plane passing through the point (x_1, y_1, z_1) is given by

$$a(x-x_1) + b(y-y_1) + c(z-z_1) = 0 \quad \text{——— (1)}$$

where a, b, c are the drs of the normal of the plane.

Here,

$$\left.\begin{aligned} a &= 5-(-2) = 7 \\ b &= 3-(-6) = 9 \\ c &= -1-(4) = -5 \end{aligned}\right\}$$

Now, from (1) we have

$$7(x-0) + 9(y-0) - 5(z-0) = 0$$

$$\Rightarrow 7x + 9y - 5z = 0$$

Problem:

Find the acute angle between the planes $3x+2y-5z-4=0$ and $2x-3y+5z-8=0$.

Soln:

Given, $3x+2y-5z-4=0 \quad \text{——— (1)}$

& $2x-3y+5z-8=0 \quad \text{——— (2)}$

Let θ be the angle bet^n the planes (1) & (2)

Now,

$$\cos\theta = \left| \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\cdot\sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$$

$$= \left| \frac{3(2) + 2(-3) - 5(5)}{\sqrt{9+4+25}\cdot\sqrt{4+9+25}} \right|$$

$$= \left| \frac{6-6-25}{38} \right|$$

$$= \frac{25}{38}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{25}{38}\right) \approx 48.86^\circ$$

Ans)

Problem:

Show that the planes $3x - 4y + 5z = 0$ and $2x - y - 2z = 0$ are perpendicular to each other.

Soln:

Given, $3x - 4y + 5z = 0$ ——— (1)

$2x - y - 2z = 0$ ——— (2)

The two given planes will be perpendicular to each other if $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Now,

$$a_1a_2 + b_1b_2 + c_1c_2 = 3(2) + (-4)(-1) + 5(-2)$$
$$= 6 + 4 - 10$$
$$= 0$$

Hence, the planes are perpendicular to each other.

(Showed)

Problem:

Show that the planes $6x - 8y + 10z = 0$ and $3x - 4y + 5z = 7$ are parallel to each other.

Soln:

Given, $6x - 8y + 10z = 0$ ——— ①

$3x - 4y + 5z = 7$ ——— ②

The given planes will be parallel if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Now, $\frac{a_1}{a_2} = \frac{6}{3} = 2$

$\frac{b_1}{b_2} = \frac{-8}{-4} = 2$ & $\frac{c_1}{c_2} = \frac{10}{5} = 2$

Hence, the planes are parallel to each other.

(Showed)

Problem:

Find the equation of a plane that passes through the point (1, 2, 3) and is parallel to the plane $3x+4y-5z=0$

Soln:

Given, $3x+4y-5z=0$ ——— (1)

We know that the eqn of the plane which is parallel to (1) is given by

$3x+4y-5z+k=0$ ——— (2) which passes through the point (1, 2, 3).

Hence, from (2) we have

$3(1)+4(2)-5(3)+k=0$

$\Rightarrow 3+8-15+k=0$

$\Rightarrow k=4$

Substituting the value of k into (2) to obtain

$3x+4y-5z+4=0$

which is the required eqn of the plane.

End

Problem: (Final 2022)

Find the equation of the plane which passes through the points $(2,2,1)$ and $(9,3,6)$ and is perpendicular to the plane $2x+6y+6z+9=0$

Soln:

We know that the eqn of the plane that passes through the points (x_1,y_1,z_1) is given by

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0 \quad \text{——— (1)}$$

Now, (1) passes through the point $(2,2,1)$

$$a(x-2)+b(y-2)+c(z-1)=0 \quad \text{——— (2)}$$

Also, plane (2) passes through the point $(9,3,6)$

Therefore,

$$a(9-2)+b(3-2)+c(6-1)=0$$

$$\Rightarrow 7a+b+5c=0 \quad \text{——— (3)}$$

Again, the plane $2x+6y+6z+9=0$ is perpendicular to the plane (2).

Hence,

$$2a+6b+6c=0 \quad \text{——— (4)}$$

Now, eliminating a, b, c from (2), (3), (4) we get.

$$\begin{vmatrix} x-2 & y-2 & z-1 \\ 7 & 1 & 5 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

$\Rightarrow (x-2)(6-30) - (y-2)(42-10) + (z-1)(42-2) = 0$

$\Rightarrow -24(x-2) - 32(y-2) + 40(z-1) = 0$

$\Rightarrow 6(x-2) + 8(y-2) - 10(z-1) = 0$

$\Rightarrow 3(x-2) + 4(y-2) - 5(z-1) = 0$

$\Rightarrow 3x + 4y - 5z - 6 - 8 + 5 = 0$

$\Rightarrow 3x + 4y - 5z = 9.$ //

Ans)

Problem:

Find the equation of the plane passing through the intersection line of the planes $x - y + 2z - 3 = 0$ and $2x - y - 3z = 0$ and the point $(4, -3, 2)$.

Soln:

Given, $x - y + 2z - 3 = 0$ ——— (1)

$2x - y - 3z = 0$ ——— (2)

So, the eqn of the plane passing through the intersection line of the planes ① & ② is given by

$$x-y+2z-3+k(2x-y-3z)=0 \quad \text{——— ③}$$

which passes through the given point $(4,-3,2)$

Thus,

$$4+3+4-3+k(8+3-6)=0$$

$$\Rightarrow 8+k(5)=0$$

$$\Rightarrow k=-8/5$$

So, from ③ we obtain

$$x-y+2z-3-\frac{8}{5}(2x-y-3z)=0$$

$$\Rightarrow 5x-5y+10z-15-16x+8y+24z=0$$

$$\Rightarrow 11x-3y-34z+15=0$$

which is the required eqn of the plane

(Ans)

Problem : (Final 2021)

Find the eqn of the plane passing through the line of intersection of the plane $2x-y=0$ and $3z-y=0$ and perpendicular to the plane $4x+5y-3z+7=0$.

Soln: Given, $2x-y=0$ ——— ①

$3z-y=0$ ——— ②

& $4x+5y-3z+7=0$ ——— (3)

We know the equⁿ of the plane passing through the line of intersection of the planes (1) & (2) is given by

$$2x-y+k(3z-y)=0$$

$$\Rightarrow 2x-(k+1)y+3kz=0 \quad \text{——} \quad (4)$$

Now, when the planes (4) & (3) are perpendicular to each other then

$$4(2)+5(-(k+1))-3(3k)=0$$

$$\Rightarrow 8-5(k+1)-9k=0$$

$$\Rightarrow 8-5k-5-9k=0$$

$$\Rightarrow 14k=3$$

$$\Rightarrow k=3/14$$

Substituting the value of k into (4) to obtain

$$2x-(3/14+1)y+3(\tfrac{3}{14})z=0$$

$$\Rightarrow 2x-\frac{17}{14}y+\frac{9}{14}z=0$$

$$\Rightarrow 28x-17y+9z=0$$

which is the required equⁿ of the plane.

Problem:

Find the equation of the line of intersection of the planes $2x-y+2z+7=0$ and $x+2y-3z+6=0$.

Soln:

Let l, m, n be the dc's of the line of intersection of the given planes.

Since the line of intersection is at right angles to the normals to both the planes, so, we have

$$\left.\begin{aligned} 2l-m+2n &= 0 \\ l+2m-3n &= 0 \end{aligned}\right\} \cdots \text{(1)}$$

Solving (1) we get.

$$\frac{l}{3-4} = \frac{m}{2+6} = \frac{n}{4+1}$$

$$\Rightarrow \frac{l}{-1} = \frac{m}{8} = \frac{n}{5}$$

$$\Rightarrow \frac{l}{1} = \frac{m}{-8} = \frac{n}{-5} \qquad \text{(2)}$$

Let $(x', y', 0)$ be a point on the line of intersection of the two given planes. Then its eqn is

$$\frac{x-x'}{l} = \frac{y-y'}{m} = \frac{z-0}{n} \qquad \text{(3)}$$

Taking the help of eqn (2), (3) can be written as

$$\frac{x-x'}{1} = \frac{y-y'}{-8} = \frac{z}{-5} \quad \text{——— (4)}$$

Again the point $(x', y', 0)$ lies on both the given planes.

$$\left.\begin{aligned} 2x'-y'+7=0 \\ x'+2y'+6=0 \end{aligned}\right\}$$

Solving the system we get.

$$x' = -4 \quad \& \quad y' = -1$$

Now, substituting the values of x' & y' into (4) we obtain

$$\frac{x+4}{1} = \frac{y+1}{-8} = \frac{z}{-5}$$

which is the required eqn of the line.

~~Ans~~

Problem: (Final 2024)

Find the equations of the planes through the points (2, 3, 1), (4, -5, 3) and parallel to the coordinate axes.

Soln:

Let the equation of the plane parallel to x-axis be

$$ay + bz + c = 0 \quad \text{——— (1)}$$

which passes through the points (2,3,1) & (4,-5,3)

Therefore,

$$3a+b+c=0$$
$$\& \quad -5a+3b+c=0$$

Eqn of any plane:
$ax+by+cz+d=0$
$\vec{n} = a\hat{i}+b\hat{j}+c\hat{k} = \langle a,b,c\rangle$ is the normal vector of the plane & is $\perp$ to the plane.
Dr's of x-axis are (1,0,0)

$$\frac{a}{-2} = \frac{b}{-8} = \frac{c}{14} = k \text{ (say)}$$

$\therefore a=-2k, \; b=-8k \; \& \; c=14k$

For a given plane to be parallel to x-axis is
$\langle a,b,c\rangle . \langle 1,0,0\rangle = 0$
$\Rightarrow \boxed{a=0}$

From (1) we get,

$$-2ky-8kz+14k=0$$
$$\Rightarrow y+4z-7=0 \quad \text{Ans}$$

Again, the eqn of the plane parallel to y-axis be

$$ax+bz+c=0 \qquad \text{——— (2)}$$

which passes through the points (2,3,1) & (4,-5,3)

Therefore,

$$2a+b+c=0$$
$$\& \quad 4a+3b+c=0$$

$$\frac{a}{-2} = \frac{b}{2} = \frac{c}{2} = k, \text{ (say)}$$

$\therefore a=-2k, \; b=2k \; \& \; c=2k$

from (2) we get,

$$-2kx + 2kz + 2k = 0$$

$$\Rightarrow x - z - 1 = 0$$

Ans

finally, let the eqn of the plane prallel to z-axis be

$$ax + by + c = 0 \quad \text{——— (3)}$$

which passes through the points (2,3,1) & (4,-5,3)

Therefore,

$$\left.\begin{aligned} 2a + 3b + c &= 0 \\ 4a - 5b + c &= 0 \end{aligned}\right\}$$

Solving. $\frac{a}{8} = \frac{b}{2} = \frac{c}{-22} = k$, (say)

$\therefore a = 8k,\ b = 2k$ & $c = -22k$.

(3) $\Rightarrow$

$$8kx + 2ky - 22k = 0$$

$$\Rightarrow 4x + y - 11 = 0$$

Ans

Problem:

Find the equations of the two planes passing through the points (0,4,-3) and (6,-4,3) which cuts off from the axes intercepts if the sum of their intercepts on the three axes is zero.

whose sum is zero.

Soln:

Let the equⁿ of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \text{——— (1)}$$

which passes through the points $(0, 4, -3)$ & $(6, -4, 3)$

Therefore, we have

$$\frac{4}{b} - \frac{3}{c} = 1 \quad \text{——— (2)}$$

& $$\frac{6}{a} - \frac{4}{b} + \frac{3}{c} = 1 \quad \text{——— (3)}$$

(2) + (3)

$$\frac{6}{a} = 2$$

$$\Rightarrow a = 3$$

From (2) we get ~~write~~

$$\frac{4}{b} = 1 + \frac{3}{c} = \frac{3+c}{c}$$

$$\Rightarrow \frac{4c}{b} = (c+3) \Rightarrow 4c = bc + 3b$$

$$\Rightarrow 4c - 3b = bc \quad \text{——— (4)}$$

A.T.Q $a + b + c = 0$

$$\Rightarrow b = -c - a = -c - 3$$

④ ⇒

$$4c = b(-c-3) + 3(-c-3)$$

$$\Rightarrow 4c = -c^2 - 3c - 3c - 9$$

$$\Rightarrow c^2 + 10c + 9 = 0$$

$$\Rightarrow c^2 + 9c + c + 9 = 0$$

$$\Rightarrow c(c+9) + 1(c+9) = 0$$

$$\Rightarrow (c+1)(c+9) = 0$$

$\therefore c = -1$ & $c = -9$

$\therefore b = 1 - 3 = -2$ (when $c = -1$)

& $b = 9 - 3 = 6$ (when $c = -9$)

Thus substituting these values into ① to obtain

$$\frac{x}{3} + \frac{y}{-2} + \frac{z}{-1} = 1$$

$$\Rightarrow \frac{x}{3} - \frac{y}{2} - \frac{z}{1} = 1$$

$$\Rightarrow 2x - 3y - 6z = 6$$

$$\frac{x}{3} + \frac{y}{6} + \frac{z}{-9} = 1$$

$$\Rightarrow \frac{x}{3} + \frac{y}{6} - \frac{z}{9} = 1$$

$$\Rightarrow 6x + 3y - 2z = 18$$

which are the required eqn's of the planes

Ans

Problem:

Show that the lines $\frac{x+1}{2} = \frac{y-2}{2} = \frac{z}{1}$ & $\frac{x-1}{6} = \frac{y+1}{1} = \frac{z-3}{5}$ are coplanar. Also find their point of intersection, and the equation of the plane containing them.

Soln:

The given lines are

$$\frac{x+1}{2} = \frac{y-2}{2} = \frac{z}{1}$$

$$\Rightarrow \frac{x-(-1)}{2} = \frac{y-2}{2} = \frac{z-0}{1} \quad \text{——— (1)}$$

and,

$$\frac{x-1}{6} = \frac{y+1}{1} = \frac{z-3}{5}$$

$$\Rightarrow \frac{x-1}{6} = \frac{y-(-1)}{1} = \frac{z-3}{5} \quad \text{——— (2)}$$

The first and second line represented by eqn's (1) & (2), respectively pass through $(-1,2,0)$ and $(1,-1,3)$ respectively and their direction ratios are $[2,2,1]$ and $[6,1,5]$. Therefore, the two lines will be coplanar, if

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1-(-1) & -1-2 & 3-0 \\ 2 & 2 & 1 \\ 6 & 1 & 5 \end{vmatrix} = 0$$

Now,

$$\begin{vmatrix} 2 & -3 & 3 \\ 2 & 2 & 1 \\ 6 & 1 & 5 \end{vmatrix} = 2(10-1)+3(10-6)+3(2-12)$$
$$= 18+12-30$$
$$= 0$$

which is true.

Hence, the given lines are coplanar.
(Showed)

To find the point of intersection, let

$$\frac{x+1}{2} = \frac{y-2}{2} = \frac{z}{1} = t_1 \text{ (say)}$$

$$\therefore \frac{x+1}{2} = t_1 \Rightarrow x = 2t_1 - 1$$

$$\frac{y-2}{2} = t_1 \Rightarrow y = 2t_1 + 2$$

$$\frac{z}{1} = t_1 \Rightarrow z = t_1$$

& $$\frac{x-1}{6} = \frac{y+1}{1} = \frac{z-3}{5} = t_2$$

$$\therefore \frac{x-1}{6} = t_2 \Rightarrow x = 6t_2 + 1$$

$$\frac{y+1}{1} = t_2 \Rightarrow y = t_2 - 1$$

$$\frac{z-3}{5} = t_2 \Rightarrow z = 5t_2 + 3$$

For point of intersection, we have

$$2t_1 - 1 = 6t_2 + 1$$

$$\Rightarrow 2t_1 = 6t_2 + 2$$

$$\Rightarrow t_1 = 3t_2 + 1 \quad \text{——— (3)}$$

Again,

$$t_1 = 5t_2 + 3$$

$$\Rightarrow 3t_2 + 1 = 5t_2 + 3 \quad [\text{using (3)}]$$

$$\Rightarrow 2t_2 = -2$$

$$\Rightarrow t_2 = -1$$

From (3) we have

$$t_1 = 3(-1) + 1 = -2$$

Therefore,

$$x = 2t_1 - 1 = 2(-2) - 1 = -5$$

$$y = 2t_1 + 2 = 2(-2) + 2 = -2$$

& $z = t_1 = -2$

Thus, the lines ① & ② intersect at $(-5, -2, -2)$

Now, the equⁿ of the plane containing the lines ① & ② is given by

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x+1 & y-2 & z \\ 2 & 2 & 1 \\ 6 & 1 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (x+1)(10-1) - (y-2)(10-6) + z(2-12) = 0$$

$$\Rightarrow 9(x+1) - 4(y-2) - 10z = 0$$

$$\Rightarrow 9x - 4y - 10z + 9 + 8 = 0$$

$$\Rightarrow 9x - 4y - 10z + 17 = 0$$

Ans

Problem:

Find the point of intersection of the line $x-1=2y=3z+2$ with the plane $5x-6y-z-4=0$

Soln:

The given equations are,

$$x-1=2y=3z+2 \quad \text{——} (1)$$

& $$5x-6y-z-4=0 \quad \text{——} (2)$$

From (1) we obtain

$$\frac{x-1}{1}=\frac{y-0}{1/2}=\frac{z-(-2/3)}{1/3}=t \text{ (say)} \quad \text{——} (3)$$

Now,

$$\left.\begin{aligned} \frac{x-1}{1}=t &\Rightarrow x=t+1 \\ \frac{y-0}{\frac{1}{2}}=t &\Rightarrow y=\tfrac{1}{2}t \\ \frac{z-(-2/3)}{1/3}=t &\Rightarrow z=\tfrac{1}{3}t-\tfrac{2}{3} \end{aligned}\right\}$$

Putting the values of x, y, z into (2), we have

$$5(t+1)-6\left(\tfrac{1}{2}t\right)-\left(\tfrac{1}{3}t-\tfrac{2}{3}\right)-4=0$$

$$\Rightarrow 5t+5-3t-\tfrac{1}{3}t+\tfrac{2}{3}-4=0$$

$$\Rightarrow 2t - \frac{1}{3}t + 1 + \frac{2}{3} = 0$$

$$\Rightarrow \frac{6t - t}{3} + \frac{3+2}{3} = 0$$

$$\Rightarrow \frac{5t}{3} + \frac{5}{3} = 0$$

$$\Rightarrow t + 1 = 0 \quad \Rightarrow t = -1.$$

Then,

$$x = t + 1 = -1 + 1 = 0$$

$$y = \frac{1}{2}t = -\frac{1}{2}$$

$$z = \frac{1}{3}t - \frac{2}{3} = \frac{1}{3}(-1) - \frac{2}{3} = -1$$

The point of intersection is $\left(0, -\frac{1}{2}, -1\right)$.

~~Ans~~

Problem: (Incourse 2022)

Find the equation of the plane through the intersection of the planes $x - 2y + z - 6 = 0$ and $2x + y - 2z - 3 = 0$ and through the point $(1, 2, 3)$. Also, find the equation of the plane through the intersection of the above-mentioned planes and perpendicular to the plane $3x + 4y - 3z - 5 = 0$.

Soln:

The given planes are,

$$x - 2y + z - 6 = 0 \quad \text{——— (1)}$$

& $$2x + y - 2z - 3 = 0 \quad \text{——— (2)}$$

Any plane through the intersection of the two planes (1) & (2) is given by

$$x - 2y + z - 6 + k(2x + y - 2z - 3) = 0 \quad \text{——— (3)}$$

$$\Rightarrow (1+2k)x + (k-2)y + (1-2k)z - (6+3k) = 0 \quad \text{——— (3)}$$

which passes through the point (1, 2, 3)

Therefore,

$$1 - 4 + 3 - 6 + k(2 + 2 - 6 - 3) = 0$$

$$\Rightarrow -6 + k(-5) = 0$$

$$\Rightarrow k = -6/5$$

Now, Substituting the value of k into (3) we obtain

$$x - 2y + z - 6 - \frac{6}{5}(2x + y - 2z - 3) = 0$$

$$\Rightarrow 5x - 10y + 5z - 30 - 12x - 6y + 12z + 18 = 0$$

$$\Rightarrow -7x - 16y + 17z - 12 = 0$$

$$\Rightarrow 7x + 16y - 17z + 12 = 0$$

Ans

Again, we need to find the equation of the plane which is through the intersection of ① & ② is perpendicular to the plane

$$3x+4y-3z-5=0 \quad \text{——} \quad (4)$$

Hence, the required cond^n would be

$$3(1+2k)+4(k-2)-3(1-2k)=0$$

$$\Rightarrow 3+6k+4k-8-3+6k=0$$

$$\Rightarrow 16k-8=0$$

$$\Rightarrow k=1/2$$

Substituting the value of k into ③ we obtain

$$\left(1+1\right)x+\left(\frac{1}{2}-2\right)y+\left(1-1\right)z-\left(6+\frac{3}{2}\right)=0$$

$$\Rightarrow 2x-\frac{3}{2}y-\frac{15}{2}=0$$

$$\Rightarrow 4x-3y-15=0$$

Ans)

Problem:

Find the equation of the plane through the point $(2,-1,-4)$ and perpendicular to the planes $3x+4y-5z+6=0$ and $x-2y+2z+1=0$.

Soln:

The equation of any plane passing through the point $(2,-1,-4)$ is given by

$$a(x-2)+b(y+1)+c(z+4)=0 \quad \text{---------} (1)$$

The equation (1) is perpendicular to each of the planes $3x+4y-5z+6=0$ & $x-2y+2z+1=0$

Thus, we have

$$\left.\begin{aligned} 3a+4b-5c &= 0 \\ \& \quad a-2b+2c &= 0 \end{aligned}\right\}$$

By cross multiplication, we get.

$$\frac{a}{8-10} = \frac{b}{-5-6} = \frac{c}{-6-4}$$

$$\Rightarrow \frac{a}{-2} = \frac{b}{-11} = \frac{c}{-10}$$

$$\Rightarrow \frac{a}{2} = \frac{b}{11} = \frac{c}{10} = k \text{ (say)}$$

$$\therefore a = 2k, \quad b = 11k \quad \& \quad c = 10k$$

Substituting the values of a, b, c into (1) to obtain

$$2k(x-2)+11k(y+1)+10k(z+4)=0$$

$$\Rightarrow 2x-4+11y+11+10z+40=0$$

$$\Rightarrow 2x+11y+10z+47=0$$ Ans

Example: (Incourse 2022)

Find the distance of the plane $7x+y+2z-16=0$ from the point $(1,1,-2)$ measured parallel to the line $\frac{x}{3}=\frac{y}{2}=\frac{z}{-4}$.

Soln:

The given plane is $7x+y+2z-16=0$ ——— (1)

and the line is $\frac{x}{3}=\frac{y}{2}=\frac{z}{-4}$ ——— (2)

The equation of the line passing through the point $(1,1,-2)$ and parallel to the given line (2) is

$$\frac{x-1}{3}=\frac{y-1}{2}=\frac{z+2}{-4}=m \text{ (say)}$$

$$\therefore \frac{x-1}{3}=m \Rightarrow x=3m+1$$

$$\frac{y-1}{2}=m \Rightarrow y=2m+1$$

$$\frac{z+2}{-4}=m \Rightarrow z=-4m-2 \quad \text{——— (3)}$$

Putting the values of x, y, z into (1) to obtain

$$7(3m+1)+2m+1+2(-4m-2)-16=0$$

$$\Rightarrow 21m+7+2m+1-8m-4-16=0$$

$$\Rightarrow 15m-12=0$$

$\Rightarrow m = 4/5$

Therefore,

$$x = 3(4/5) + 1 = 17/5$$

$$y = 2(4/5) + 1 = 13/5$$

$$z = -4(4/5) - 2 = -\frac{26}{5}$$

From (3), the coordinates of the point of intersection are $\left(\frac{17}{5}, \frac{13}{5}, -\frac{26}{5}\right)$

The required distance $= \sqrt{\left(\frac{17}{5} - 1\right)^2 + \left(\frac{13}{5} - 1\right)^2 + \left(-\frac{26}{5} + 2\right)^2}$

$$= \sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{8}{5}\right)^2 + \left(-\frac{16}{5}\right)^2}$$

$$= \sqrt{\frac{12^2 + 8^2 + 16^2}{5^2}}$$

$$= \frac{\sqrt{464}}{5}$$

$$= \frac{4\sqrt{29}}{5}$$

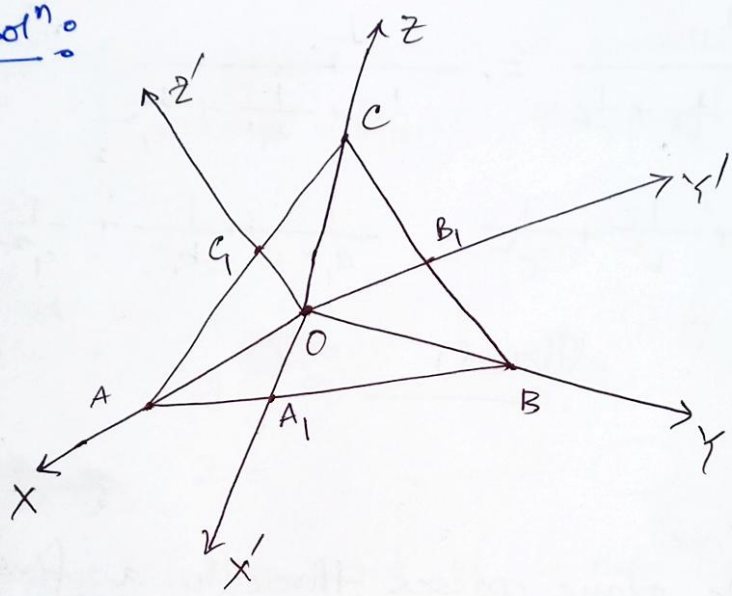
Ans

Problem:

Two systems of rectangular axes have the same origin. If a plane cuts them at distances a, b, c and a_1, b_1, c_1, respectively from the origin, prove that

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2}.$$

Proof/soln:



$OA = a$
$OB = b$
$OC = c$

$OA_1 = a_1$
$OB_1 = b_1$
$OC_1 = c_1$

Let OX, OY, OZ and OX', OY', OZ' be the two system of axes. According to the question, the equation of the plane with reference to OX, OY, OZ is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \text{——— (1)}$$

And the equ^n of the plane with reference to OX', OY', OZ' is

$$\frac{x}{a_1} + \frac{y}{b_1} + \frac{z}{c_1} = 1 \quad \text{——— (2)}$$

Since the planes ① & ② are the same planes. So the lengths of perpendicular to ① & ② from the origin will mean the length of the same perpendicular

$$\frac{1}{\sqrt{\left(\frac{1}{a}\right)^2+\left(\frac{1}{b}\right)^2+\left(\frac{1}{c}\right)^2}} = \frac{1}{\sqrt{\left(\frac{1}{a_1}\right)^2+\left(\frac{1}{b_1}\right)^2+\left(\frac{1}{c_1}\right)^2}}$$

$$\Rightarrow \frac{1}{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}} = \frac{1}{\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c^2}}$$

$$\Rightarrow \frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2} = \frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c^2}$$

(Proved)

Problem :

A variable plane passes through a fixed point (a,b,c) and meets the axes at P, Q, R. Show that the locus of the point of intersection of the planes through P, Q, R parallel to the coordinate plane is $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$.

Soln :

Let the equation of the variable plane be

$$\frac{x}{a_1}+\frac{y}{b_1}+\frac{z}{c_1}=1 \quad \text{——— ①}$$

As the plane ① passes through the point (a, b, c)

$$\therefore \frac{a}{a_1} + \frac{b}{b_1} + \frac{c}{c_1} = 1 \quad \text{——} (2)$$

The plane ① cuts the coordinate axes at P, Q, and R, so the coordinates of P, Q and R are respectively $P(a_1, 0, 0)$, $Q(0, b_1, 0)$ & $R(0, 0, c_1)$.

The eqⁿ of the plane which is parallel to the yz-plane and passes through the point $P(a_1, 0, 0)$ is given by

$$x = a_1 \quad \text{——} (3)$$

Similarly, $y = b_1$ —— (4)

& $z = c_1$ —— (5)

The locus of the point of intersection of these planes ③, ④ & ⑤ is obtained by eliminating the parameters a_1, b_1, c_1 between the eqⁿ ③, ④, ⑤.

Putting the values of a_1, b_1, c_1 from ③, ④, ⑤ in ②, we obtain the required locus, which is given by

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$$

(Proved)

Problem:

Find the length and equation of the shortest distance (SD) between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$

Soln:

The given lines are

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{——— (1)}$$

$$\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5} \quad \text{——— (2)}$$

Let l, m, n be the 'dc's of the SD line. Since SD line is perpendicular to both lines (1) and (2), some have

$$\left.\begin{aligned} 2l + 3m + 4n &= 0 \\ 3l + 4m + 5n &= 0 \end{aligned}\right\}$$

By cross-multiplication, we get

$$\frac{l}{15-16} = \frac{m}{12-10} = \frac{n}{8-9}$$

$$\Rightarrow \frac{l}{-1} = \frac{m}{2} = \frac{n}{-1}$$

$$\Rightarrow \frac{l}{-1} = \frac{m}{2} = \frac{n}{-1} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{(-1)^2+(2)^2+(-1)^2}} = \frac{1}{\sqrt{6}}$$

$$\therefore \quad l = -\frac{1}{\sqrt{6}}, \quad m = \frac{2}{\sqrt{6}}, \quad n = -\frac{1}{\sqrt{6}}$$

The lines ① & ② passes through the point (1,2,3) and (2,4,5), respectively.

SD = Projection of the line joining the two points (1,2,3) and (2,4,5) on the SD line

$$= l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

$$= -\frac{1}{\sqrt{6}}(2-1) + \frac{2}{\sqrt{6}}(4-2) - \frac{1}{\sqrt{6}}(5-3)$$

$$= -\frac{1}{\sqrt{6}} + \frac{4}{\sqrt{6}} - \frac{2}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{6}} \quad \text{Ans}$$

Now, the eqn of the plane which contains the line ① and the SD line is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ -1 & 2 & -1 \end{vmatrix} = 0$$

$$\Rightarrow (x-1)(-3-8) - (y-2)(-2+4) + (z-3)(4+3) = 0$$

$$\Rightarrow -11(x-1) - 2(y-2) + 7(z-3) = 0$$

$$\Rightarrow -11x - 2y + 7z + 11 + 4 - 21 = 0$$

$\Rightarrow 11x + 2y - 7z + 6 = 0$ Ans

Again, the equⁿ of the plane which contains the line ② and the SD line is given by

$$\begin{vmatrix} x-2 & y-4 & z-5 \\ 3 & 4 & 5 \\ -1 & 2 & -1 \end{vmatrix} = 0$$

$\Rightarrow (x-2)(-4-10) - (y-4)(-3+5) + (z-5)(6+4) = 0$

$\Rightarrow -14(x-2) - 2(y-4) + 10(z-5) = 0$

$\Rightarrow -14x - 2y + 10z + 28 + 8 - 50 = 0$

$\Rightarrow 14x + 2y - 10z + 14 = 0$

$\Rightarrow 7x + y - 5z + 7 = 0$

Ans

Problem:

Find the equation of the straight line that intersects the lines $4x + y - 10 = 0 = y + 2z + 6$, & $3x - 4y + 5z + 5 = 0 = x + 2y - 4z + 7$ and passes through the point $(-1, 2, 2)$.

Solⁿ:

Given, the two lines are,

$4x+y-10=0=y+2z+6$ ——— (1)

& $3x-4y+5z+5=0=x+2y-4z+7$ ——— (2)

Any plane containing line (1) is given by

$$4x+y-10+\lambda_1(y+2z+6)=0$$

$$\Rightarrow 4x+(1+\lambda_1)y+2\lambda_1 z+(-10+6\lambda_1)=0 \quad \text{——— (3)}$$

Again, any plane containing line (2) is given by

$$3x-4y+5z+5+\lambda_2(x+2y-4z+7)=0$$

$$\Rightarrow (3+\lambda_2)x+(-4+2\lambda_2)y+(5-4\lambda_2)z+(5+7\lambda_2)=0 \quad \text{——— (4)}$$

Since the required line passes through the intersections of the given lines, it is possible by choosing λ_1 and λ_2 properly to make this line coplanar with the planes represented by (3) & (4) and hence to obtain its equation as the intersection of these two planes.

Since, the line passes through the point $(-1,2,2)$.

Therefore from (3) & (4) we have

$$4(-1)+(1+\lambda_1)2+2\lambda_1\cdot 2+(-10+6\lambda_1)=0$$

$$\Rightarrow -4+2+2\lambda_1+4\lambda_1-10+6\lambda_1=0$$

$$\Rightarrow 12\lambda_1=12 \quad \boxed{\Rightarrow \lambda_1=1}$$

Also,

$$(3+\lambda_2)(-1)+(-4+2\lambda_2)(2)+(5-4\lambda_2)(2)+5+7\lambda_2=0$$

$$\Rightarrow -3-\lambda_2-8+4\lambda_2+10-8\lambda_2+5+7\lambda_2=0$$

$$\Rightarrow 2\lambda_2+4=0 \Rightarrow \boxed{\lambda_2=-2}$$

Substituting λ_1 & λ_2 in (3) & (4), the eqn of the required line is given by

$$4x+2y+2z-4=0$$

$$\Rightarrow 2x+y+z-2=0$$

and $\quad x-8y+13z-9=0$

or, $\quad 2x+y+z-2=0=x-8y+13z-9 \quad \text{——— (5)}$

or, in the symmetrical form by

$$\frac{x+1}{21}=\frac{y-2}{-25}=\frac{z-2}{-17}.$$

Ans

(Plane-related Problems)

Problem:

Find the equation of the plane which is perpendicular to the plane $5x+3y+6z+8=0$ and contains the line of intersection of the planes $x+2y+3z-4=0$ and $2x+y-z+5=0$.

Solution:

Given,

$$x+2y+3z-4=0 \quad \text{——} (1)$$

$$2x+y-z+5=0 \quad \text{——} (2)$$

So the eqn of the plane passing through the intersection line of the planes (1) & (2) is given by

$$x+2y+3z-4+k(2x+y-z+5)=0 \quad \text{——} (3a)$$

$$\Rightarrow (1+2k)x+(2+k)y+(3-k)z-4+5k=0 \quad \text{——} (3b)$$

Since (3b) is perpendicular to the plane $5x+3y+6z+8=0$

So we have

$$(1+2k)\cdot 5+(2+k)3+(3-k)\cdot 6=0$$

$$\Rightarrow 5+10k+6+3k+18-6k=0$$

$$\Rightarrow 7k+29=0$$

$$\Rightarrow 7k=-29$$

$$\therefore k=-\frac{29}{7}$$

Substituting the value of k into (3a), we obtain

$$x+2y+3z-4+\left(-\frac{29}{7}\right)(2x+y-z+5)=0$$

$$\Rightarrow 7x+14y+21z-28-58x-29y+29z-145=0$$

$$\Rightarrow -51x-15y+50z-173=0$$

$$\Rightarrow 51x+15y-50z+173=0$$

which is the required eqn of the plane.

Done

Problem:

Find the equation of a plane passing through the line of intersection of the planes $x-2y+3z+4=0$ and $2x-3y+4z-1=0$ and is perpendicular to the plane $3x-y+2z-1=0$

Solution:

The equation of any plane which passes through the line of intersection of the two planes

$$x-2y+3z+4=0$$

& $2x-3y+4z-1=0$ is given by

$$x-2y+3z+4+k(2x-3y+4z-1)=0 \quad \text{——} (1)$$

$\Rightarrow (1+2k)x + (-2-3k)y + (3+4k)z + 4 - k = 0$ — (2)

Since the plane (2) is perpendicular to the given plane $3x - y + 2z - 1 = 0$, so we have

$$3(1+2k) - 1(-2-3k) + 2(3+4k) = 0$$

$\Rightarrow 3 + 6k + 2 + 3k + 6 + 8k = 0$

$\Rightarrow 17k = -11$

$\Rightarrow k = -\frac{11}{17}$

Putting the value of k into eqn (1) we have

$$x - 2y + 3z + 4 - \frac{11}{17}(2x - 3y + 4z - 1) = 0$$

$\Rightarrow 17x - 34y + 51z + 68 - 22x + 33y - 44z + 11 = 0$

$\Rightarrow -5x - y + 7z + 79 = 0$

$\Rightarrow 5x + y - 7z - 79 = 0$

which is the required eqn of the plane.

Ans

Problem:

Find the equation of the plane through the point $(1,-4,2)$ and perpendicular to the two planes $2x+5y-z-12=0$ and $4x-7y+3z+8=0$.

Solution:

We know that the equation of the plane that passes through the point (x_1, y_1, z_1) is given by

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$$

$$\therefore \quad a(x-1)+b(y+4)+c(z-2)=0 \qquad \text{---------- (1)}$$

where (a,b,c) are the drs of the plane (1).

ATQ, the plane (1) is perpendicular to the planes $2x+5y-z-12=0$ & $4x-7y+3z+8=0$

$$\therefore \quad 2a+5b-c=0 \qquad \text{---------- (2)}$$

$$\& \quad 4a-7b+3c=0 \qquad \text{---------- (3)}$$

Using cross-multiplication technique, we get from (2) & (3)

$$\frac{a}{15-7} = \frac{b}{-4-6} = \frac{c}{-14-20}$$

$$\Rightarrow \frac{a}{8} = \frac{b}{-10} = \frac{c}{-34}$$

$$\Rightarrow \frac{a}{4} = \frac{b}{-5} = \frac{c}{-17} = K \text{ (say)}$$

$\therefore a = 4K,\ b = -5K,\ \&\ c = -17K$

Substituting the values of a, b, c into ① we get

$$4K(x-1) - 5K(y+4) - 17K(z-2) = 0$$

$$\Rightarrow 4x - 4 - 5y - 20 - 17z + 34 = 0$$

$$\Rightarrow 4x - 5y - 17z + 10 = 0$$

which the equⁿ of the required plane.

Problem:

Find the equation of the plane perpendicular to each of the planes $x - 4y + z = 0$ and $3x + 4y + z - 2 = 0$ at a distance unity from the origin.

Solⁿ:

Given,

$$x - 4y + z = 0 \quad \text{——— ①}$$

$$3x + 4y + z - 2 = 0 \quad \text{——— ②}$$

Let the required equ^{n} of the plane be

$$ax+by+cz+d=0 \quad \text{---------} (3)$$

Since plane (3) is perpendicular to the planes (2) & (1), therefore

$$\left.\begin{aligned} a-4b+c&=0 \\ 3a+4b+c&=0 \end{aligned}\right\} \quad \text{---------} (4)$$

Solving the system (4) using cross-multiplication, we get

$$\frac{a}{-8}=\frac{b}{2}=\frac{c}{16}=k \text{ (say)}$$

$\therefore a=-8k,\ b=2k,\ c=16k$

Substituting the values of a, b, c into (3) we get

$$-8kx+2ky+16kz+d=0$$

$$\Rightarrow 4x-y-8z=\frac{d}{2k} \quad \text{---------} (5)$$

ATQ,

$$\left|\frac{\frac{d}{2k}}{\sqrt{4^2+(-1)^2+(-8)^2}}\right|=1$$

$$\Rightarrow \left| \frac{\frac{d}{2K}}{\sqrt{16+1+64}} \right| = 1$$

$$\Rightarrow \left| \frac{d/2K}{\sqrt{81}} \right| = 1$$

$$\Rightarrow \frac{d}{18K} = 1$$

$$\left.\begin{array}{l} \Rightarrow K = d/18 \\ \Rightarrow d = 18K \end{array}\right\} \text{or}$$

Substituting the value of d into (5) we get

$$4x - y - 8z = \frac{18K}{2K} = 9$$

$$\Rightarrow 4x - y - 8z = 9$$

which is the required eqn of the plane.